

LECTURE 8:

Cooperation in MAS (IV): voting protocols

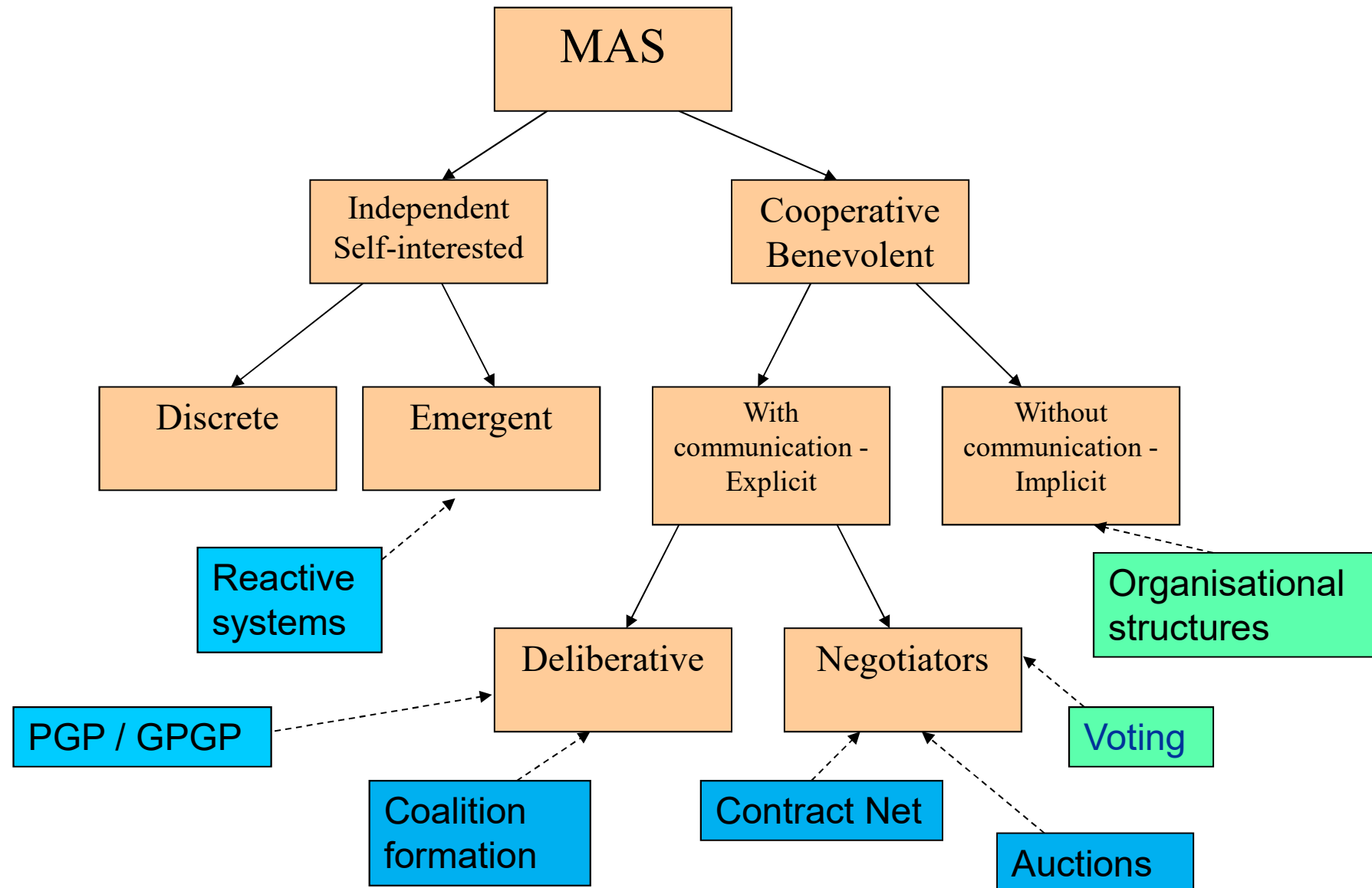
Introduction to Multi-Agent Systems (MESIIA, MIA)

Department of Computer Science and Mathematics

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Fall 2019

Cooperation hierarchy [last lectures]



Outline of the lecture

- Voting protocols

- Social choice
 - Properties of social choice rules
 - Simple voting protocols
 - Plurality / Anti-plurality / Best-Worst / Approval
 - Total order protocols: Binary / Borda / Condorcet
 - Complex voting mechanisms
 - Linguistic votes
 - Uncertain opinions
-

Distributed Decision Making - Voting

- Mechanism which chooses the outcome of a **negotiation** based on the inputs (**votes**) given by all agents to a set of competing options
 - We will assume that voters are **truthful** (they vote for the candidate they think is best)
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Insincere (Strategic) Voters

- Self-interested agents can benefit from *insincerely* declaring their preferences
 - Suppose your choice will likely come in second place. If you rank the first choice of the rest of the group (insincerely) very low, you may lower that choice enough so yours is first
 - “*Useful vote*” effect in politics: more than two candidates, and you are quite certain that your preferred candidate doesn’t have any chance
 - However, knowledge of the true preferences of all voters is rarely available
-

Basic elements in a voting protocol

- **Aim** of the negotiation: rank **a set of alternatives** based on the individual ranking of those options by each agent
 - A - set of n agents
 - O - set of m alternatives
 - Each agent i has a **preference relation**
 $<_i : O \times O$
-

Preference relations

- **Weak order:** complete and transitive

R_1	R_2
x_1	$x_2 \quad x_4$
$x_2 \quad x_3$	x_3
x_4	x_1

- **Linear order:** weak order +
anti-symmetric

R_3
x_3
x_1
x_2
x_4

Social choice rule

- **Input:** the agents preference relations
($<_1, \dots, <_n$)
 - **Output:** elements of O **sorted** according to the input
(**social preference relation** $<^*$ of the agent group)
 - In other words, it creates an **ordering of the group of alternatives**, so that the most (**socially**) preferred alternative is chosen
 - This order may be **partial**
-

Desirable properties of the social choice rule (I)

- **Calculability**

A social preference ordering $<^*$ should **exist** for all possible inputs

- **Completeness**

$<^*$ should be defined for every pair of alternatives $(o, o') \in O$

- **Linearity**

$<^*$ should be **antisymmetric** and **transitive** over O

Desirable properties of the social choice rule (II)

- **Anonymity / No dictatorship**

The outcome of the social choice rule depends on the set of opinions, but not on which agents have these opinions.

No agent i should be a *dictator* in the sense that $o <_i o'$ implies $o <^* o'$ regardless of the preferences of the other agents

Desirable properties of the social choice rule (III)

- Unanimity / Pareto efficiency

If $\forall i \in A (o <_i o')$ then $(o <^* o')$

Do not misorder the options if all agents agree.

If everybody thinks that A is better than B, A should be preferred to B in the aggregated order.

Desirable properties of the social choice rule (IV)

- **Neutrality**

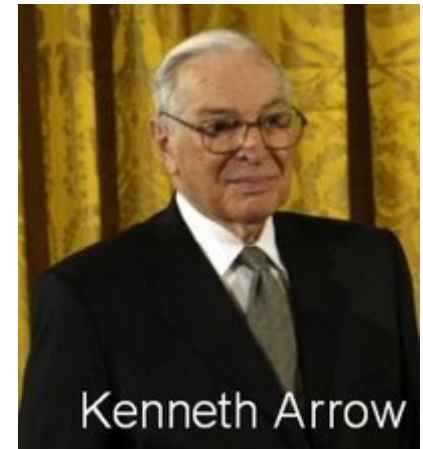
The outcome of the social choice rule should not depend on how alternatives are named or ordered

- **Independence of irrelevant alternatives**

Removing / Adding an irrelevant alternative should not affect the winner of the vote

Arrow's impossibility theorem-1951

- No social choice rule satisfies all of these conditions
 - E.g. in many cases an irrelevant alternative affects the outcome of the voting protocol
- Some of the constraints must be relaxed
 - We may not require $<^*$ to be always defined
 - We may not require that $<^*$ is asymmetric and transitive



Economy
Nobel Prize
1972

(youngest, 51)

Simple voting mechanisms

- Some basic voting mechanisms
 - Plurality / Anti-plurality / Best-Worst / Approval
 - Protocols based on total orders
 - Binary protocol (series of votes of 2 options each)
 - Borda protocol (sum of all the preferences of the agents)
 - Condorcet protocol (pairwise comparison of options, given full preference ordering of each agent)
 - All the protocols are problematic in one sense or another
-



Plurality protocol

- Each agent can give 1 vote to 1 of the alternatives
- The alternative with the highest number of votes wins

Vote for one option.

☐	Joe Smith
☒	John Citizen
☐	Jane Doe
☐	Fred Rubble
☐	Mary Hill

Problems of plurality voting (I)

■ *Useful vote*

- ❑ Quite common effect of political polls
 - ❑ 45 % option A, 40% option B, 15% option C
 - ❑ The most socially preferred option is A => it should be the winner if all agents vote truthfully
 - ❑ The agents that prefer option C know that they have no possibility to win, and most of them (80%) prefer option B to option A => 80% vote for B and 20% for A
 - A: 48% B: 52%, and B finally wins
-

Problems of plurality voting (II)

- Huge effect of irrelevant alternative
 - 2 basic options, A and B, around 50% each
 - Another –very minority option- C appears, attracting 1% of the voters of A
 - A: 49% B:50% C:1% and B wins
 - Scarce information about the preferences of each voter
 - Each agent can only give 1 vote, even if it considers 2-3-4 “good” alternatives
-

Problems of plurality voting (III)

- Strange effects

- 42% A C B D
 - 26% B C D A
 - 15% C D B A
 - 17% D C B A
 - A wins with 42% of the votes
 - 58% of the voters preferred other options
 - 58% consider A the **worst** option
 - 100% consider C the 1st-2nd best option
-

Advantages of plurality voting

- Most **simple** voting mechanism
 - Very **efficient** from the computational point of view
 - **Equality** principle, as it preserves the idea of **1 agent = 1 vote**
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Another version: anti-plurality

- Each voter gives a **negative** vote to the alternative he considers the **worst**
 - The option with **less** votes wins
 - In the previous example, A would have 58 negative votes!
-

Anti-plurality example

- 30% CBDA
- 30% CADB
- 20% ABDC
- 20% BADC

Last: C (40% negative votes) – but also first option for 60%

A and B get 30% negative votes

D is the winner with 0 negative votes – but it was not the first or second option for anyone

Best-worst voting systems

- Each agent gives a **positive** vote to his best alternative and a **negative** vote to his worst alternative
 - Each alternative receives $\alpha > 0$ points for each positive vote and $-\delta < 0$ points for each negative vote
 - The option with more points wins
-

Example

	1	2	3	4	5	6	7
	C	C	B	B	A	A	A
	B	B	C	C	B	C	C
	A	A	A	A	C	B	B

$A \rightarrow 3\alpha - 4\delta$ points $B \rightarrow 2\alpha - 2\delta$ points $C \rightarrow 2\alpha - \delta$ points

Plurality ($\delta = 0$) $A \succ B \sim C$

Anti-plurality ($\alpha = 0$) $C \succ B \succ A$

$\alpha = \delta = 1$ $C \succ B \succ A$

$\alpha = 2$ and $\delta = 1$ $C \succ A \sim B$

$\alpha = 4$ and $\delta = 1$ $A \succ C \succ B$

Approval voting

- Each voter selects a subset of the candidates
 - The candidate with most votes wins
 - **k-approval voting**
 - Each voter selects a subset of k candidates
 - $k=1$: plurality
 - $k=n-1$: anti-plurality
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Protocols based on linear orders

- Each voter gives a **full list of the options**, ordered according to his preferences (from best to worst)
 - A voter prefers option i to option j if option i appears before option j in his list
-

Binary protocol

- All the options are **ordered** and then **evaluated in pairs** (options 1 and 2, the winner with option 3, the winner with option 4, etc.)
- **Simple majority:** option A is better than option B if and only if the number of voters that prefer A to B is greater than the number of voters that prefer B to A
- The option that wins the last evaluation is the overall winner

$\text{win}(o_5, \text{win}(o_4, \text{win}(o_3, \text{win}(o_2, o_1))))$

The ordering problem

- $x > z > y$ (35%)
 $y > x > z$ (33%)
 $z > y > x$ (32%)
 - Note that y is preferred to x (65-35), x is preferred to z (68-32), and z is preferred to y (67-33)
 - $\text{win}(x, \text{win}(y, z)) = x$
 - $\text{win}(y, \text{win}(x, z)) = y$
 - $\text{win}(z, \text{win}(x, y)) = z$
 - The order of the pairings affects the outcome !
 - The voter organiser may influence the result
 - The last options have more chances of winning
 - No Neutrality
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Another problematic example

35% of agents have preferences $c > d > b > a$

33% of agents have preferences $a > c > d > b$

32% of agents have preferences $b > a > c > d$

Evaluation in the order *acbd*:

Win(a,c)=a Win(a,b)=b Win(b,d)=d $\Rightarrow$ d Wins

- d was the worst alternative for 32%
- d was not the best alternative for anyone
- Everybody prefers c to d (!) – No Unanimity

Summary of problems of binary voting

- Decisive role of the **ordering** of the alternatives
 - An alternative x may win even if there is another alternative x' which is preferred to x by all agents
 - Alternatives **may be misordered**
 - **Temporal cost** of the voting process (sequence of pairwise eliminative votes)
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Borda protocol-1770

- For each voter, we assign $|O|$ points to the preferred option, $|O|-1$ points to the second, and so on
 - Example: 4 possibilities, one agent considers the order ABCD
A:4 points, B:3 points, C:2 points, D:1 point



Borda protocol result

- The points are added across the voters and the alternative with the highest count becomes the social choice
[similar to Eurovision song contest]



Borda paradox

- $a > b > c > d$

- $b > c > d > a$

- $c > d > a > b$

- $a > b > c > d$

- $b > c > d > a$

- $c > d > a > b$

- $a > b > c > d$

$a=18, b=19, c=20, d=13$

If the worst alternative –d–
is removed

- $a > b > c$

- $b > c > a$

- $c > a > b$

- $a > b > c$

- $b > c > a$

- $c > a > b$

- $a > b > c$

$a=15, b=14, c=13$

Even if we keep the relative preferences between a, b and c, the final result changes completely

Inverted-order paradox

- *Borda protocol* with 4 alternatives

$x=22$, $a=17$, $b=16$, $c=15$

Order: xabc

1. $x > c > b > a$
2. $a > x > c > b$
3. $b > a > x > c$
4. $x > c > b > a$
5. $a > x > c > b$
6. $b > a > x > c$
7. $x > c > b > a$

- If we remove x:

$c=15$, $b=14$, $a=13$

The second one is c, not a

1. $c > b > a$
2. $a > c > b$
3. $b > a > c$
4. $c > b > a$
5. $a > c > b$
6. $b > a > c$
7. $c > b > a$

Problems of the Borda protocol

- Most computationally expensive
 - The Eurovision Song Contest is endless ...
 - Eliminating (or adding) one irrelevant alternative may totally change the outcome of the protocol
 - Winner => Last
 - Second worst => Winner
 - Total order changes if options are removed one by one
-

Borda protocol with weak orders

- The Borda protocol has been extended to manage **weak orders** in different ways
- A simple one: an option o receives from a voter v as many points as the number of **options that are considered worst than o by v**

$\frac{R_1}{x_1}$	$\frac{R_2}{x_2 \quad x_4}$	$\frac{R_3}{x_3}$
$x_2 \quad x_3$	x_3	x_1
x_4	x_1	x_2
		x_4
$x_1 \rightarrow 3 + 0 + 2 = 5$	$x_2 \rightarrow 1 + 2 + 1 = 4$	
$x_3 \rightarrow 1 + 1 + 3 = 5$	$x_4 \rightarrow 0 + 2 + 0 = 2$	
$x_1 \sim x_3 \succ x_2 \succ x_4$		

Condorcet protocol-1785

- Each voter ranks the candidates in order of preference
- Each candidate is compared to each other
- If a candidate wins **all** the comparisons, it is the winner of the election
- In the event of a tie, use another resolution method (e.g. Borda count)



Example (I)



- Election of the capital city of Tennessee
- Everybody prefers to have the capital as close as possible

42% of voters (close to Memphis)	26% of voters (close to Nashville)	15% of voters (close to Chattanooga)	17% of voters (close to Knoxville)
1. Memphis	1. Nashville	1. Chattanooga	1. Knoxville
2. Nashville	2. Chattanooga	2. Knoxville	2. Chattanooga
3. Chattanooga	3. Knoxville	3. Nashville	3. Nashville
4. Knoxville	4. Memphis	4. Memphis	4. Memphis

Example (II)

Pair	Winner
Memphis (42%) vs. Nashville (58%)	Nashville
Memphis (42%) vs. Chattanooga (58%)	Chattanooga
Memphis (42%) vs. Knoxville (58%)	Knoxville
Nashville (68%) vs. Chattanooga (32%)	Nashville
Nashville (68%) vs. Knoxville (32%)	Nashville
Chattanooga (83%) vs. Knoxville (17%)	Chattanooga

		A			
		Memphis	Nashville	Chattanooga	Knoxville
B	Memphis		[A] 58% [B] 42%	[A] 58% [B] 42%	[A] 58% [B] 42%
	Nashville	[A] 42% [B] 58%		[A] 32% [B] 68%	[A] 32% [B] 68%
	Chattanooga	[A] 42% [B] 58%	[A] 68% [B] 32%		[A] 17% [B] 83%
	Knoxville	[A] 42% [B] 58%	[A] 68% [B] 32%	[A] 83% [B] 17%	
Ranking:		4th	1st	2nd	3rd

Nashville wins

Problem of Condorcet method

- Possibility of **circular ambiguities**
 - No alternative wins to all the other alternatives
 - There are many ways to resolve them
 - Keep the candidate that **wins more matches** (Copeland)
 - Take into account the **relative strengths of defeats** (Minimax, Ranked Pairs, Schulze, ...)
 - You can look at the winning votes or at the winning margin
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Generalization of voting protocols

- Scoring vector

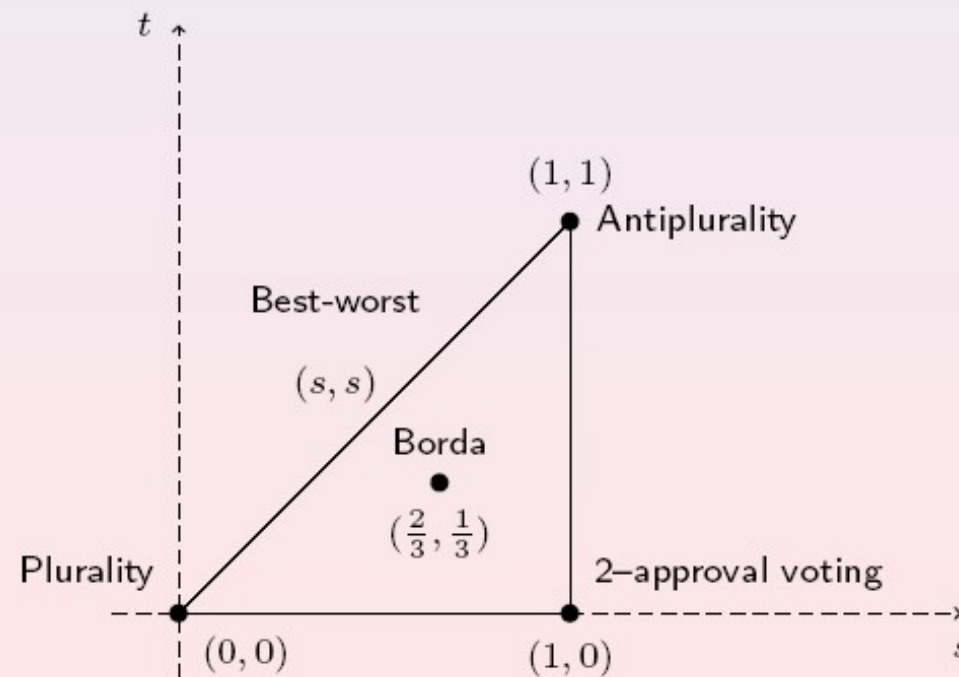
$S=(s_1, \dots, s_n)$ with $s_i \geq s_{i+1}$ and $s_1 > s_n$

- Each voter gives s_i points to his i -th option
 - The option with more points wins
 - Examples with 4 options
 - vector $(4,3,2,1) \Rightarrow$ Borda protocol
 - vector $(1,0,0,0) \Rightarrow$ Plurality protocol
 - vector $(0,0,0,-1) \Rightarrow$ Anti-plurality protocol
 - vector $(1,0,0,-1) \Rightarrow$ Best-worst
-

Normalization of scoring rules

$$(s_1, \dots, s_n) \equiv \left(1, \frac{s_2 - s_n}{s_1 - s_n}, \dots, \frac{s_{n-1} - s_n}{s_1 - s_n}, 0\right)$$

$$n = 4: \{(1, s, t, 0) \mid 0 \leq t \leq s \leq 1\} \equiv \{(s, t) \in [0, 1]^2 \mid t \leq s\}$$



Example of influence of protocol (I)

- 45 ABCD, 25 CDBA, 20 DBCA, 10 DCBA
 - Plurality protocol: A-45 B-0 C-25 D-30
 - A wins quite clearly, B has no votes
 - However, note that 55% say that A is the worst option (!)
-

Example of influence of protocol (II)

- 45 ABCD, 25 CDBA, 20 DBCA, 10 DCBA
 - Pairwise comparisons
 - A loses with B,C and D
 - B wins C, C wins D, D wins B
 - Binary protocol
 - A can never win (!)
 - Order ABCD: D wins
 - Order BDAC: C wins
 - Order CDBA: B wins
-

Example of influence of protocol (III)

- 45 ABCD, 25 CDBA, 20 DBCA, 10 DCBA
 - Condorcet protocol: no option beats all the others, and B/C/D tied with 2 victories
 - Borda protocol
 - A: 235 B: 265 C:260 D:240
 - B wins (although all the options get similar values)
 - B is not the first option for anyone !!!
 - B is a good consensual option (2nd for 65%, 3rd for 35%), whereas A was very problematic (45% very positive support, but 55% very negative support)
 - Recall that B had 0 votes in the plurality protocol (!)
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More complex voting mechanisms

- Use of **linguistic information** to represent the opinion of each voter with respect to each alternative
 - Management of **uncertainty** in the voter's opinions
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Use of linguistic information

- Each voter can use a **linguistic label** from a predetermined set to evaluate each alternative

R	To Reject
P	Poor
A	Acceptable
G	Good
VG	Very Good
E	Excellent

	1	2	3	4	5	6	7
Alan	A	R	VG	E	A	P	A
Benny	VG	E	G	VG	G	G	G
Charles	G	P	E	R	G	G	VG

Evaluation

- The evaluation of each of the alternatives is also a linguistic label, obtained with an **aggregation** of the voter's opinions
 - Aggregator function
 - Idempotent
 - Monotonic
 - Example: the result of the aggregation of $A = \{a_1, \dots, a_n\}$ can be the label l_i that minimizes the distance($A, \{l_i, \dots, l_i\}$)
-

Distance measure

- The distance between two vectors of linguistic labels may be defined as the addition of the distances between the labels in each position
 - $\text{Distance}(\text{Poor}, \text{Poor}) = 0$
 - $\text{Distance}(\text{Poor}, \text{Good}) = 2$
 - $\text{Distance}(\text{Reject}, \text{Excellent}) = 5$

R	To Reject
P	Poor
A	Acceptable
G	Good
VG	Very Good
E	Excellent

Example

R	To Reject
P	Poor
A	Acceptable
G	Good
VG	Very Good
E	Excellent

	1	2	3	4	5	6	7
Alan	A	R	VG	E	A	P	A
Benny	VG	E	G	VG	G	G	G
Charles	G	P	E	R	G	G	VG

	R	P	A	G	VG	E	Global evaluation
Alan	16	11	8	11	14	19	Acceptable
Benny	25	18	11	4	5	10	Good (Very Good)
Charles	19	14	11	8	11	16	Good

Uncertain preferences

- Each voter represents the opinion on every alternative with an **interval** defined over an ordered set of linguistic labels

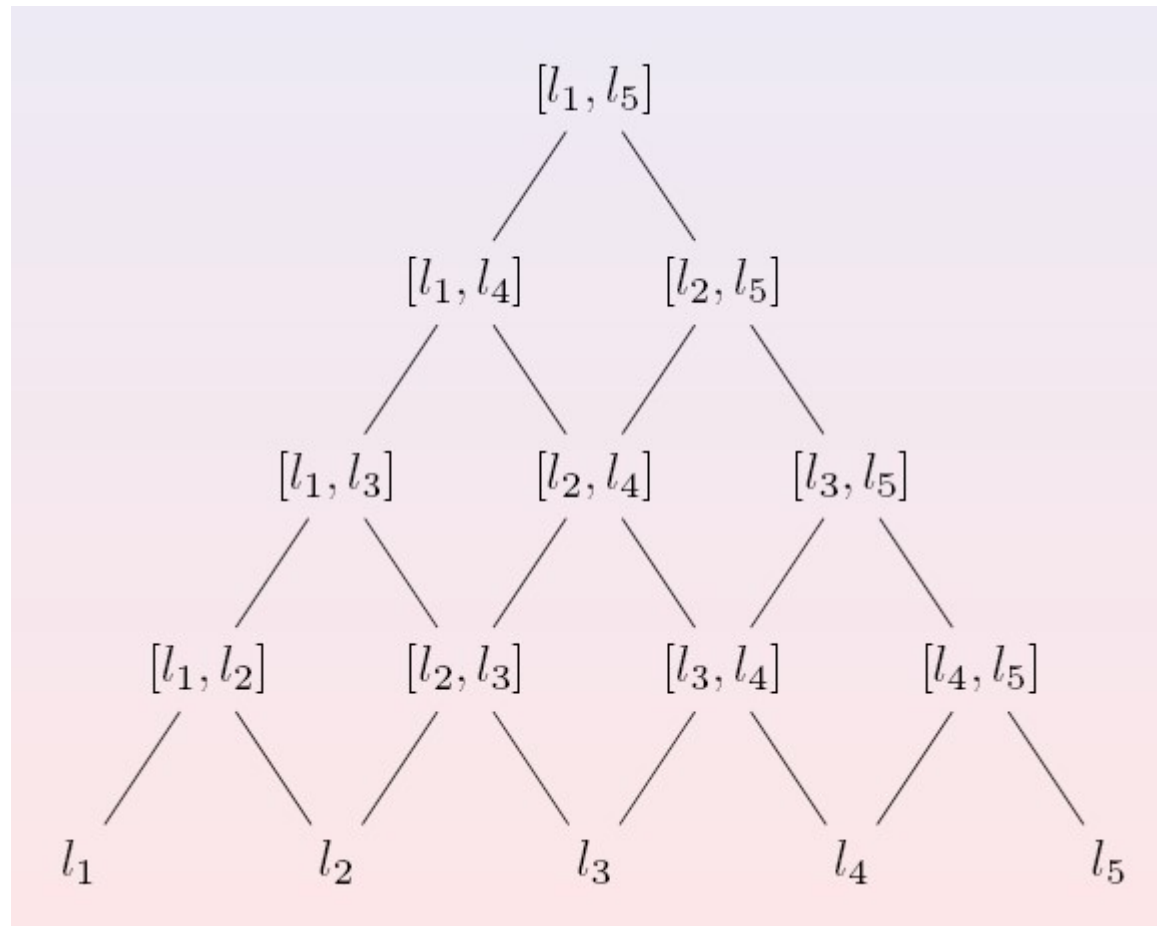
l_1	l_2	l_3	l_4	l_5
very bad	bad	acceptable	good	very good
$[l_4, l_5]$ means <i>between good and very good</i>				

Graph representation of intervals (5 labels)

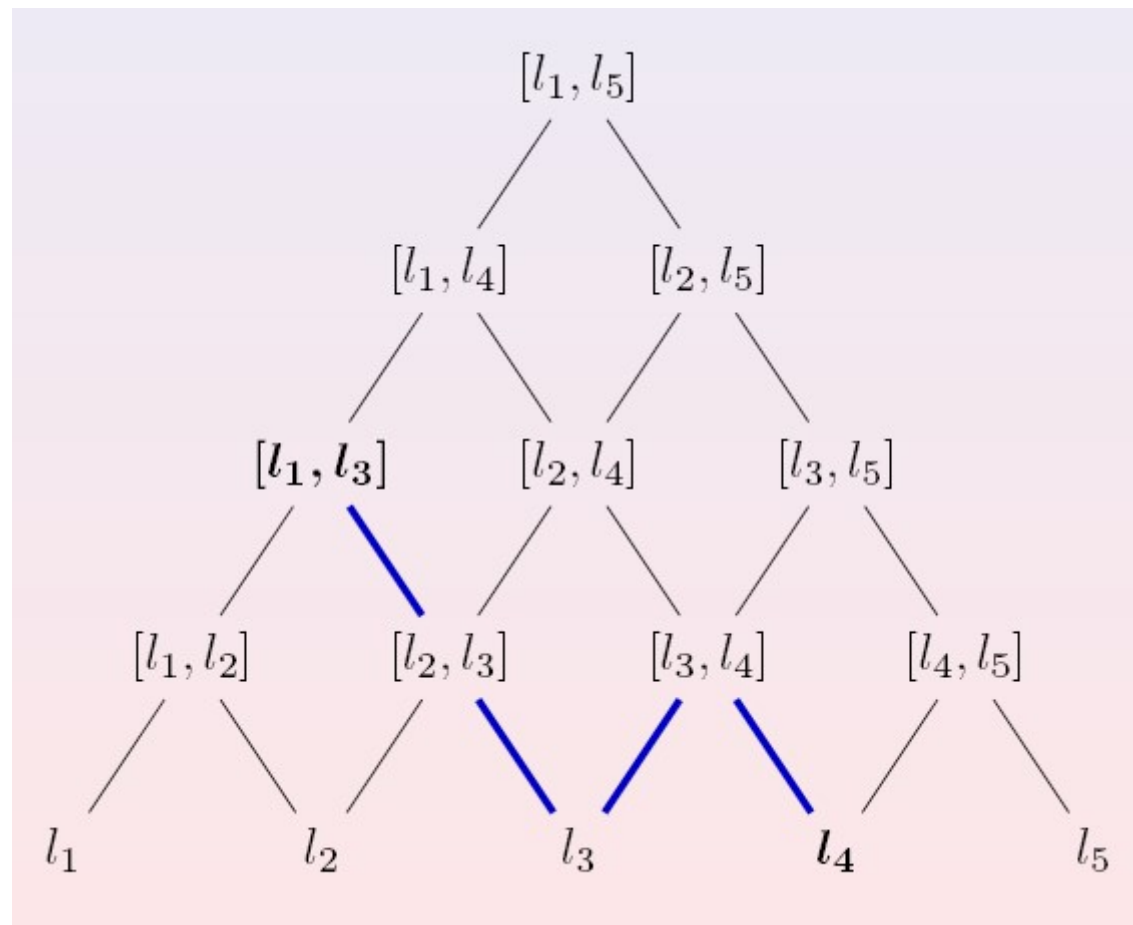
*More general,
imprecise*



*More specific,
precise*



Distance between linguistic intervals



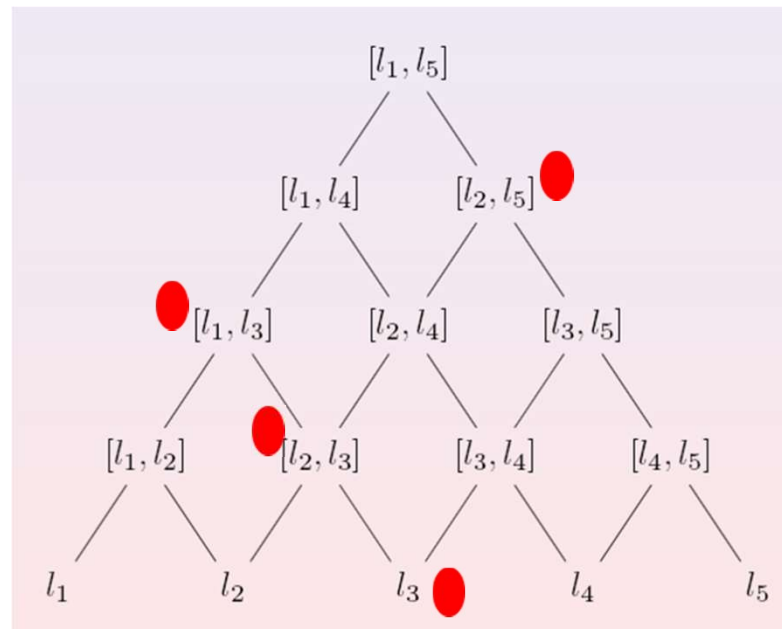
Evaluation of an alternative

- Given m opinions (linguistic intervals) on a certain item x_i , the result of the aggregation of the m opinions may be defined as the interval $/$ that minimizes the addition of the distances from $/$ to each of the opinions.

$$E(x_i) = \left\{ \mathcal{E} \in \mathbb{L} \mid \forall \mathcal{F} \in \mathbb{L} \quad \sum_{p=1}^m d(\mathcal{E}, v_i^p) \leq \sum_{p=1}^m d(\mathcal{F}, v_i^p) \right\}$$

Example

	Agent 1	Agent 2	Agent 3	Agent 4	Joint opinion
Option A	(A, VG)	(G, G)	(A, G)	(B, G)	(A, G)
Option B	(A, A)	(VB, A)	(B, A)	(B, VG)	(B, A)
Option C	(B, G)	(B, A)	(VG, VG)	(B, G)	(B, G)



l_1	l_2	l_3	l_4	l_5
very bad	bad	acceptable	good	very good

Readings for this week

- Chapter 12 of the book by M.Wooldridge “An introduction to Multi-Agent Systems” (2nd edition).
 - Chapter 9 of the book by M.Fasli “Agent technology for e-commerce”.
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